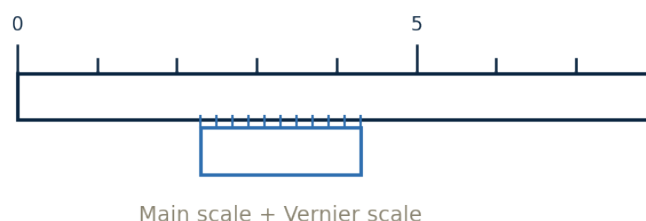


# Units & Measurements

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

## 1. Notes

- Fundamental (base) quantities in SI: length (m), mass (kg), time (s), electric current (A), temperature (K), amount of substance (mol), luminous intensity (cd).
- Dimensional formula expresses a quantity in powers of [M], [L], [T]. Used to check the correctness of equations (principle of homogeneity) and to derive relations between physical quantities.
- Significant figures indicate the precision of a measurement. Rules: all non-zero digits are significant; zeros between non-zero digits are significant; trailing zeros after a decimal point are significant.
- Error types: systematic (instrumental, zero error - can be corrected) and random (unpredictable, reduced by repeated measurement and averaging).
- Least count = smallest value an instrument can measure. Vernier caliper LC = 1 MSD - 1 VSD. Screw gauge LC = pitch / number of divisions on circular scale.
- Relative (fractional) error =  $\Delta a / a$ . Percentage error =  $(\Delta a / a) \times 100$ . For a quantity  $Z = A^p B^q / C^r$ , the maximum fractional error is  $p(\Delta A/A) + q(\Delta B/B) + r(\Delta C/C)$ .



*Vernier caliper: main scale + vernier scale for precise length measurement.*

## 2. Formula Sheet

|                                    |                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------|
| Dimensional formula of force       | $[F] = [M L T^{-2}]$                                                                         |
| Dimensional formula of energy/work | $[E] = [M L^2 T^{-2}]$                                                                       |
| Dimensional formula of pressure    | $[P] = [M L^{-1} T^{-2}]$                                                                    |
| Vernier caliper least count        | $LC = 1 \text{ MSD} - 1 \text{ VSD}$                                                         |
| Screw gauge least count            | $LC = \text{Pitch} / \text{No. of circular divisions}$                                       |
| Max % error in $Z = A^p B^q / C^r$ | $\% \text{error}(Z) = p(\% \text{error } A) + q(\% \text{error } B) + r(\% \text{error } C)$ |
| Relative error                     | $\Delta a / a_{\text{mean}}$                                                                 |

## 3. Practice Problems (22 questions)

- The SI unit of coefficient of viscosity is (a)  $\text{N s/m}^2$  (b)  $\text{N/m}^2$  (c)  $\text{N m/s}$  (d)  $\text{N/m}$ . Identify the correct option.
- Find the dimensional formula of Planck's constant  $h$ , given  $E = h \times \text{frequency}$ .
- Find the dimensional formula of gravitational constant  $G$ , given  $F = G m_1 m_2 / r^2$ .
- A physical quantity  $Z = A^2 B^3 / C$ . If the percentage errors in  $A$ ,  $B$ ,  $C$  are 1%, 2%, 3% respectively, find the maximum percentage error in  $Z$ .
- The length of a rod is measured as 25.0 cm using a scale with least count 0.1 cm. Express the measurement with correct significant figures and its absolute error.
- In a vernier caliper, 1 MSD = 1 mm and 10 VSD = 9 MSD. Find the least count of the instrument.
- A screw gauge has a pitch of 1 mm and 100 divisions on the circular scale. Find its least count.
- How many significant figures are there in 0.00580?
- Round off 3.14159 to 3 significant figures.
- The density of a cube is measured by measuring its mass and the length of its side. If the maximum error in mass is 4% and in length is 3%, find the maximum error in density.
- Check the dimensional correctness of the equation  $v = u + at$ .
- Find the dimensions of angular momentum.
- The value of Stefan's constant is  $5.67 \times 10^{-8} \text{ W/(m}^2 \text{ K}^4)$ . Find its dimensional formula.
- Two resistors of resistance  $R_1 = 100 \pm 3 \text{ ohm}$  and  $R_2 = 200 \pm 4 \text{ ohm}$  are connected in series. Find the equivalent resistance with its absolute error.
- Convert a speed of 72 km/h into m/s.
- If the unit of force is 100 N, unit of length is 10 m, and unit of time is 100 s, find the unit of mass in this system.
- Name two derived quantities that have the same dimensional formula as work.

18. A student measures the diameter of a wire using a screw gauge and gets readings 1.20 mm, 1.22 mm, 1.21 mm, 1.23 mm. Find the mean value and the mean absolute error.
19. What is the order of magnitude of the number of atoms in 1 gram of hydrogen?
20. State the principle of homogeneity of dimensions and give one use of dimensional analysis.
21. Find the number of significant figures in the measured value  $6.320 \times 10^4$  kg.
22. Distinguish between accuracy and precision of a measurement with one example each.
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## Answer Key

- |                                                              |                                                                                            |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1. (a) $\text{N s/m}^2$                                      | 2. $[\text{M L}^2 \text{T}^{-1}]$                                                          |
| 3. $[\text{M}^{-1} \text{L}^3 \text{T}^{-2}]$                | 4. 16%                                                                                     |
| 5. $25.0 \pm 0.1$ cm (3 significant figures)                 | 6. 0.1 mm                                                                                  |
| 7. 0.01 mm                                                   | 8. 3                                                                                       |
| 9. 3.14                                                      | 10. 10%                                                                                    |
| 11. Dimensionally consistent: both sides $[\text{L T}^{-1}]$ | 12. $[\text{M L}^2 \text{T}^{-1}]$                                                         |
| 13. $[\text{M T}^{-3} \text{K}^{-4}]$                        | 14. $300 \pm 7$ ohm                                                                        |
| 15. 20 m/s                                                   | 16. 1 kg (new unit) = 1000 old-system kg (derive via $F=ma$ )                              |
| 17. Torque and energy (both $[\text{M L}^2 \text{T}^{-2}]$ ) | 18. Mean = 1.215 mm, mean abs. error = 0.0125 mm approx 0.01 mm                            |
| 19. Order $10^{23}$                                          | 20. Conceptual - see notes above                                                           |
| 21. 4                                                        | 22. Conceptual - accuracy = closeness to true value; precision = repeatability of readings |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.